BEYOND HIGH PRICES: AUDITING INFORMATION-MEDIATED ENFORCEMENT IN RECURRENT PRICING AGENTS

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ABSTRACT

High prices and correlated responses do not prove that pricing agents have learned collusive enforcement. We study this distinction in a continuous-price duopoly with noisy public signals and recurrent agents trained by proximal policy optimization. We use two benchmarks. A two-state trigger model shows when deviations are deterred by future losses. A Gaussian-monitoring model shows how noise weakens this threat. We audit the learned GRU policies using counterfactual price interventions and signal ablation. Across ten seeds and one million interactions, normalized price elevation drops from 0.43 in early training to 0.015 under informative monitoring, and to 0.019 after signal ablation. The paired difference is 0.0037, with a 95 percent bootstrap interval of [0.0241, 0.0288]. Crucially, losses from forced price cuts hit immediately. They do not stem from adverse future values, and several individual deviations remain profitable. Rival responses are small under informative monitoring and vanish entirely under ablation. These results separate the illusion of coordination from actual dynamic enforcement. We find no evidence of real trigger enforcement in this setting.

1 INTRODUCTION

High prices do not reveal how they are sustained. That is the plain fact. A pricing algorithm may charge more because it is still exploring. Its function approximator may carry a bias. Or deviations may invite a costly response. Only the last explanation describes the dynamic enforcement mechanism we study here. A credible claim must distinguish these possibilities. It must say which one it means.

Our economic idea is simple. An observation channel can affect the enforceability of coordinated pricing while demand and costs remain fixed. The relevant theoretical quantity is the difference between detecting a deviation and falsely punishing cooperation, multiplied by the continuation value at risk. We make this statement precise for a fixed public trigger strategy. It is a conditional result about incentives. It is not a prediction that a particular learning algorithm will discover that strategy.

The computational question is narrower. Do recurrent PPO agents learn behavior whose price elevation and response to a unilateral cut depend on an action-sensitive public signal? We compare an informative channel with a fully action-independent channel. We retrain both firms from scratch at equal training budgets. We freeze the resulting policies and measure current and continuation profit changes under paired interventions. This separates immediate pricing incentives from subsequent adjustment.

The completed experiment does not establish information-mediated enforcement. Prices approach the one-shot Nash reference in both conditions. Removing the channel does not yield a statistically resolved reduction in final prices or an increase in welfare. The largest average losses from deviation are already present in the intervention period. Moreover, a zero impulse response under action-independent monitoring is implied by the execution architecture. It cannot independently demonstrate that a cartel has collapsed.

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Our contribution is a claim-aligned combination of a transparent incentive benchmark, a counterfactual audit with a proved negative control, and a complete ten-seed PPO comparison. The Gaussian benchmark explains what enforcement would require. The experiment tests for behavioral evidence of that mechanism and reports its limits. Neither the theory nor the audit is a general characterization of algorithmic collusion.

2 Market and Observation Model

Two firms simultaneously choose prices in $[1,\, 2.5]$. Marginal cost is 1, product quality is 2, outside utility is 0, market size is 1, and the logit scale is 0.25. These primitives are identical across treatments. For firm i, demand, profit, and total surplus are

$$q_i(p_i, p_j) = \frac{\exp\big((v - p_i)/\mu\big)}{1 + \exp\big((v - p_i)/\mu\big) + \exp\big((v - p_j)/\mu\big)}, \tag{1}$$

$$\pi_i(p_i, p_j) = (p_i - c)\, q_i(p_i, p_j), \tag{2}$$

$$\mathrm{TS}(p_i, p_j) = (v - c)\big[q_i(p_i, p_j) + q_j(p_i, p_j)\big]. \tag{3}$$

The simulator computes symmetric Nash and joint-profit references on a $1{,}501$-point price grid, yielding p^{N} and p^{J}. An independent continuous calculation gives 1.472927 and 1.924981, respectively. This numerical check concerns the reference prices, not equilibrium of the learned policies. Price elevation is

$$E = \frac{\bar{p} - p^{\mathrm{N}}}{p^{\mathrm{J}} - p^{\mathrm{N}}}$$

and is not clipped to $[0, 1]$. It measures a position between reference prices. It is not a collusion probability.

2.1 Gaussian Monitoring and Posterior Contexts

At each episode reset, a joint draw θ is sampled uniformly from the retained monitoring posterior and held fixed for the episode. The implementation imposes $\theta \in \Theta$. Prices determine the latent signal mean through

$$m_t = \theta_{\mathrm{lo}} + (\theta_{\mathrm{hi}} - \theta_{\mathrm{lo}}) \cdot \frac{\min(p_{1,t}, p_{2,t}) - p^{\mathrm{N}}}{p^{\mathrm{J}} - p^{\mathrm{N}}}, \tag{4}$$

clamped so that $m_t \in [\theta_{\mathrm{lo}}, \theta_{\mathrm{hi}}]$.

The informative channel averages the most recent two available latent means and adds one Gaussian innovation after aggregation. It does not average two noisy signals. With σ_ε^2 denoting the innovation variance,

$$r_{t+1} = \frac{m_t + m_{t-1}}{2} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2). \tag{5}$$

A one-step report queue delays delivery. Reports enter the next observation only after the current action has been taken. Normalization is public and fixed: the delivered report is centered at 1.6, divided by 0.4, and clipped to $[-5, 5]$. Initial histories are zero and pending reports are neutral. No hidden posterior parameter is supplied to the actor.

In the ablation, the generated report is instead $r_{t+1} \sim \mathcal{N}(1.6,\, 0.4^2)$, independently of prices. Delay, observation length, economic primitives, and budgets are unchanged. This replacement is exact action independence, not a numerical approximation to infinite variance. It also changes the report's marginal scale and mean. Consequently, retraining identifies the effect of this channel replacement, not a pure information-content effect holding all input moments fixed.

The calibration likelihood models labeled cooperative and deviating signals as normals with different means and a shared standard deviation. Gaussian priors are placed on log distances of the means from the threshold and on log signal scale. The native model also estimates benchmark payoff parameters, but only the three monitoring coordinates enter PPO environments. Thus posterior randomization represents monitoring uncertainty. It does not randomize demand or costs. The interpolation in (4) is imposed structure. It is not an estimated relationship between real prices and reports.

3 LEARNING AND COUNTERFACTUAL AUDIT

Each firm has its own PPO actor and value head (Schulman et al., 2017). The input contains 32 public reports and normalized episode time. A 128-unit GRU encodes the report window. A projection combines its final state with time, followed by $\tanh$ and layer normalization. The GRU is recomputed from a fresh hidden state for each window. It is not an unbounded recurrent memory. Actors receive neither rivals' prices nor their own realized rewards as execution inputs, although each firm's rewards train its policy.

Training uses two environments, one million joint market interactions, $2{,}048$ rollout steps per environment when a full rollout fits, ten optimization epochs, minibatches of 256, learning rate 3×10^{-4}, discount 0.99, GAE parameter 0.95, PPO clipping 0.2, value coefficient 0.5, entropy coefficient 0.01, and gradient norm cap 0.5. Episodes terminate after 512 periods with no continuation bootstrap. Both firms learn simultaneously, so single-agent PPO arguments do not imply equilibrium convergence.

The actor samples a latent normal variable and applies $\tanh$ before mapping to prices. Its density includes the $\tanh$ Jacobian. This Gaussian action distribution controls exploration and is distinct from Gaussian observation noise. All reported evaluations deploy the frozen deterministic action $\mu(s)$. This is not generally the expected action under the stochastic training policy. The audit concerns this deployment rule only.

3.1 INTERVENTION OUTCOMES

For each trained policy pair, we intervene on either firm at period 64 or 256 using normalized action cuts of 0.1 or 0.3, giving eight probe types. Unclipped price reductions are 0.075 and 0.225 because the action-to-price slope is 0.75. Each probe uses 128 paired rollouts. Both branches share the posterior context and Gaussian innovations. After the single forced cut, both agents return to their frozen policies. The audit runs through period 511.

The rival impulse response is

$$\Delta p_t^{\mathrm{rival}} = p_t^{\mathrm{rival}}(\text{cut branch}) - p_t^{\mathrm{rival}}(\text{control branch}).$$

We report its signed sum after impact and the sum of the negative parts of its mean path. The latter is a descriptive downward-response area. It is not a structural measure of punishment. Continuation gain includes both firms' future adjustments and does not isolate the rival's causal contribution.

3.2 AN EXACT NEGATIVE CONTROL

Proposition 1 (Observation invariance)**.** *Suppose execution policies are frozen deterministic functions of public report histories and time. Reports are action independent, initial buffers and innovations are coupled, and actions do not change any other actor input. Then a one-period action intervention leaves every later prescribed action unchanged. Hence* $\Delta p_t^{\mathrm{rival}} = 0$ *for all* $t > t_0$ *and* $\Delta \pi^{\mathrm{cont}} = 0$ *pathwise in this stationary market.*

Proof. Before intervention the inputs coincide. At intervention the rival therefore chooses the same action. Action independence makes the next delivered report identical in the two branches, including queued reports. Histories and time remain identical. Induction gives identical subsequent observations and prescribed actions. Stationary payoffs depend only on current prices, so every subsequent payoff difference is zero. Only the intervention-period difference remains. □ □

4 EXPERIMENTAL EVIDENCE ACROSS TRAINING SEEDS

All ten PPO seeds in each condition were retrained from scratch. Each run records 100 evaluations with 16 episodes each. The requested $10{,}000$-interaction spacing is rounded to rollout boundaries, starting at $12{,}288$ interactions. Final outcomes use 64 separate evaluation episodes. Conditions share seed identifiers and evaluation-stream construction.

Price elevation averages 0.4318 and 0.4327 at the first checkpoint under informative and action-independent monitoring, respectively. At the final diagnostic checkpoint the means are 0.0151 and

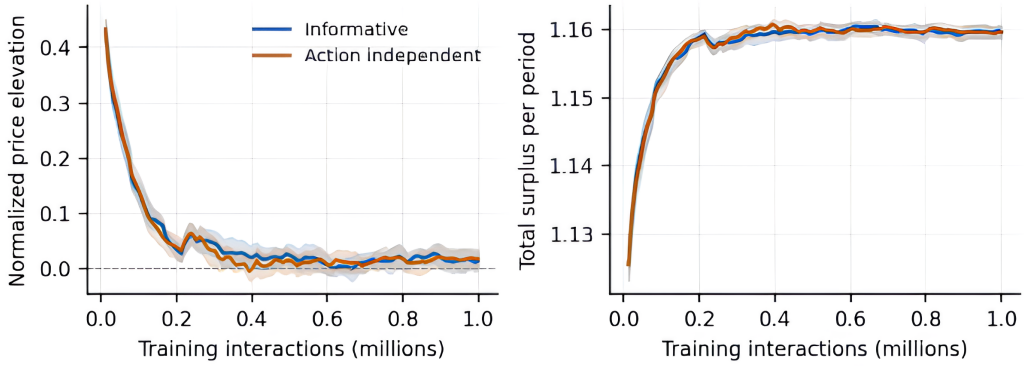


Figure 1: Deterministic deployment. Lines average ten training seeds; shading gives pointwise 95 percent intervals from 2,000 whole-seed bootstrap resamples. Checkpoints are repeated measurements. Zero elevation marks the Nash reference.

Table 1: Final evaluations. Differences are action independent minus informative. Brackets are pointwise 95 percent paired intervals from 2,000 bootstrap resamples of ten seed pairs, using seed 20260920. No multiplicity correction is applied.

Outcome	Informative	Action independent	Paired diff. [95% CI]
Mean price	1.479804	1.481492	0.001688 [−0.0109, 0.0130]
Price elevation	0.015052	0.018787	0.003735 [−0.0241, 0.0288]
Total surplus	1.159588	1.159660	0.000072 [−0.0013, 0.0017]
Feature rank (firm 0)	1.274335	1.338647	0.064312 [0.0113, 0.1337]
Feature rank (firm 1)	1.269538	1.353578	0.084040 [0.0282, 0.1495]

0.0188. Total surplus rises in both conditions. Early high prices do not distinguish the two information structures. Their subsequent decline cannot be attributed to information removal alone.

The price difference is small relative to the 0.452 Nash-to-joint-profit reference gap. But its interval includes both signs. These estimates do not establish equivalence. With only ten independent training seeds, the experiment may miss modest effects. Nor do near-Nash average prices establish approximate equilibrium. Averages can conceal asymmetric behavior and profitable deviations.

Effective rank is the exponential entropy of normalized eigenvalues of centered feature covariance. Features include time, projection, and layer normalization. They are not raw GRU states. The ablation raises rank but also changes the input distribution. Neither harmful collapse nor a causal effect of rank on pricing follows.

5 Deviation Incentives and Rival Responses

An unprofitable cut need not have been deterred. It may simply set the price too low. We therefore compare full remaining-episode gains with immediate gains, keeping the two cut sizes separate. Each seed contributes the mean over both firms and both intervention times within a cut size. Seed-level means, rather than the 128 rollouts or four probe types, are the replication unit.

The average continuation component under informative monitoring is approximately 10^{-5} for either cut. In contrast, its full gain averages -0.005192 for the smaller cut and -0.051322 for the larger. Most of the loss is already incurred at the intervention. Under the action-independent treatment, equality between immediate and full gain holds to numerical precision, as Proposition 1 predicts.

Aggregation must not conceal profitable deviations. Three small-cut conditional means are positive under informative monitoring, occurring in two training seeds; the largest is 0.003846. One is positive under action-independent monitoring, with gain 0.000122. Their stored within-policy mean-bootstrap intervals exclude zero individually, but these intervals are conditional and not ad-

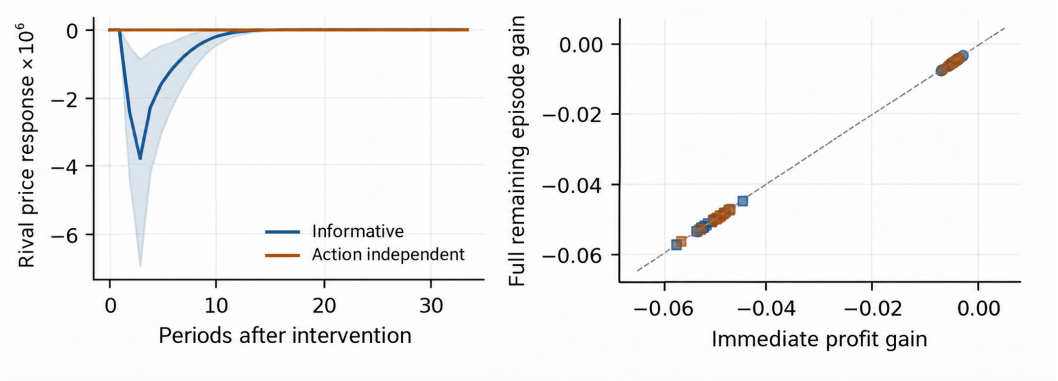


Figure 2: Left: mean rival response across eight probe types and ten seeds, with pointwise seed-bootstrap intervals, in millionths of a price unit. Right: immediate and full remaining-episode gains for each seed, averaged within cut size. Circles denote a normalized cut of 0.1; squares, 0.3. Points lie close to the equality line because continuation effects are small. The right panel does not display every individual probe.

Table 2: Price-cut audit. Entries average ten seed-level means. "Positive probes" counts positive conditional mean gains among 40 seed–firm–time combinations per cut and condition; these are not 40 independent training replications.

Channel	Action cut	Immediate gain	Full gain	Positive probes
Informative	0.1	−0.005186	−0.005192	3 of 40
Informative	0.3	−0.051317	−0.051322	0 of 40
Action independent	0.1	−0.004800	−0.004800	1 of 40
Action independent	0.3	−0.050273	−0.050273	0 of 40

justed for searching across probes. The descriptive count alone is enough to reject the statement that every tested deviation has a negative estimated gain.

The informative signed rival-response area averages -3.7×10^{-6} price-period units, with a pointwise seed-bootstrap interval of $[-1.1 \times 10^{-5}, 3.2 \times 10^{-6}]$. This detects a small response under the specified averaging rule. It does not establish economically consequential enforcement. The ablated response is exactly zero by architecture. Finally, equation (4) saturates when the minimum price is at or below the Nash reference. Different cuts can then generate identical latent signals. This limits the audit's ability to distinguish shock severity and must not be mistaken for robust punishment dynamics.

6 A Fixed Public Trigger Benchmark

We now state the theory that motivated the audit. Consider a stationary repeated stage game with discount δ, a cooperative public state C, and an absorbing punishment state P. In C, firms play a fixed action profile. A bad public signal triggers P. Punishment prescribes a stage-game Nash profile, making it credible in this stationary setting. Fix one unilateral one-period deviation followed by a return to the prescribed strategy. Write π^C, π^D, and π^P. Let α and β be bad-signal probabilities after cooperation and that deviation.

Proposition 2 (Incentive threshold)**.** *The specified deviation is unprofitable in C if and only if*

$$\delta \geq \frac{\pi^D - \pi^C}{(\pi^D - \pi^C) + (\beta - \alpha)(\pi^C - \pi^P)}. \tag{6}$$

Equivalently, $\delta \geq \delta^*$, where

$$\delta^* = \frac{g}{g + \Delta \cdot \Delta V},$$

with $g = \pi^D - \pi^C$, $\Delta = \beta - \alpha$, and $\Delta V = \pi^C - \pi^P$. A feasible discount factor below one exists exactly when $\Delta > 0$.

Proof. The prescribed and deviating values satisfy

$$V^C = \pi^C + \delta\big[(1-\alpha)V^C + \alpha V^P\big], \tag{7}$$

$$V^D = \pi^D + \delta\big[(1-\beta)V^C + \beta V^P\big]. \tag{8}$$

Subtracting (7) from (8) gives $V^D - V^C = g - \delta\Delta(V^C - V^P)$. Subtracting the punishment Bellman equation from (7) yields $V^C - V^P = \frac{\pi^C - \pi^P}{1-\delta}$. Its coefficient is strictly positive. Substitution and rearrangement give $V^D \le V^C$ if and only if $\delta \ge \delta^*$, proving (6). The denominator is positive. Requiring $\delta^* < 1$ gives the stated feasibility condition. □ □

Full cooperative-state compatibility requires the corresponding inequality for every firm and every unilateral action, not just the action with the largest current payoff. Let $g^{\max}$ denote that action's current gain, $\Delta^{\max}$ its signal probability, and ΔV the firm's cooperative–punishment payoff gap. The condition is

$$\delta \;\ge\; \frac{g^{\max}}{g^{\max} + \Delta^{\max} \cdot \Delta V} \quad \text{for every firm.} \tag{9}$$

Punishment credibility remains a separate requirement. A finite set of price-cut probes cannot verify (9) for continuous actions and every history. Some smaller-profit deviations can be harder to detect and therefore harder to deter. This is why mean deviation gain is not an equilibrium certificate.

The comparative statics follow directly. With $\Delta > 0$,

$$\frac{\partial \delta^*}{\partial \alpha} > 0, \qquad \frac{\partial \delta^*}{\partial \beta} < 0, \qquad \frac{\partial \delta^*}{\partial (\pi^C - \pi^P)} < 0. \tag{10}$$

Detection helps enforce this strategy. False alarms weaken it. Perfect monitoring gives $\alpha = 0,\ \beta = 1$. If $\Delta \le 0$, positive current gains cannot be deterred for any $\delta < 1$. These identities explain the mechanism without asserting that a trained neural policy implements it.

7 What the Gaussian Assumption Proves

For the analytical benchmark alone, let the public statistic be normally distributed with mean μ_C under cooperation, mean μ_D after the specified deviation, and common standard deviation σ. A bad signal is a statistic below a fixed threshold τ, where $\mu_D < \tau < \mu_C$. Then

$$\alpha = \Phi\left(\frac{\tau - \mu_C}{\sigma}\right), \qquad \beta = \Phi\left(\frac{\tau - \mu_D}{\sigma}\right).$$

Proposition 3 (Restricted noise comparative static)**.** *Holding the means, threshold, prescribed actions, and payoff gaps fixed, increasing σ strictly lowers $\Delta = \beta - \alpha$ and strictly raises the threshold δ^* in* (6)*.*

Proof. Set $z_C = (\tau - \mu_C)/\sigma$ and $z_D = (\tau - \mu_D)/\sigma$. With ϕ the standard normal density,

$$\frac{\partial \Delta}{\partial \sigma} = -\frac{1}{\sigma}\big[z_D\,\phi(z_D) - z_C\,\phi(z_C)\big]. \tag{11}$$

Since $\mu_D < \tau < \mu_C$, we have $z_D < 0 < z_C$, and the bracket is positive. Thus $\partial\Delta/\partial\sigma < 0$. Combining (11) with (10) gives $\partial\delta^*/\partial\sigma > 0$.

As $\sigma \to 0$, detection tends to one and false alarms to zero. As $\sigma \to \infty$, both tend to one half. The limiting threshold is 2 and is infeasible for discounted infinite-horizon play. □ □

The proposition explains one way noise can weaken enforcement. Deviations become harder to distinguish. Accidental punishment becomes more frequent. Gaussianity provides the explicit tail probabilities. It is not evidence that real monitoring errors are normal. The result also does not survive arbitrary changes in thresholds, prescribed actions, or continuation strategies without further analysis.

7.1 The Connection to PPO Is Conditional

The simulator uses the same Gaussian location-family idea but passes a continuous report to the GRU rather than enforcing the binary trigger. A posterior draw determines the signal endpoints and base noise, and equation (4) interpolates between them. Added noise enters through the variance sum in (5). The calibration prior, Gaussian signal likelihood, Gaussian innovations, and Gaussian actor exploration are distinct modeling objects. None is a Gaussian-process regression result. No Bayesian-optimization performance claim is made in this PPO study.

The present experiment compares one informative channel with one action-independent replacement. It does not vary Gaussian noise while holding a trigger strategy fixed, estimate the two error probabilities for a learned trigger, or test the derivative in (10). The baseline also aggregates and delays reports, so its learned response cannot be substituted directly into the contemporaneous two-state formula. The theorem supplies a mechanism and an incentive accounting framework. It is established by the proof above. PPO supplies finite-sample behavioral evidence.

A further distinction is the horizon. The benchmark is infinite horizon, whereas training ends after 512 periods and agents observe remaining time through normalized time. In a complete-information finite repeated game with a unique stage-game equilibrium, backward induction would select stage equilibrium play. We do not invoke that result as a theorem about these learned policies. We note that a known terminal date changes the learning problem. The experiment's near-Nash outcomes neither refute the fixed-trigger result nor verify its mechanism in neural agents.

8 Interpretation and Limits

Imperfect-monitoring theory links enforcement to information and continuation incentives (Green and Porter, 1984; Fudenberg et al., 1994). Tabular learners can sustain high prices and react to deviations in specific environments (Calvano et al., 2020). Deng et al. (2025) report near-Nash pricing under PPO and DQN. Our contribution therefore concerns channel replacement, incentive accounting, and a proved negative control, rather than the observation that PPO can learn low prices.

Under the reported market, posterior, architecture, deployment rule, and budget, the evidence does not establish that public-signal-dependent continuation losses sustain materially elevated prices. It establishes neither that PPO cannot collude nor that noise broke collusion.

Several limitations determine what would change this assessment.

Input distribution confound. Action independence changes the report distribution as well as its dependence on prices. Moment-matched replacement, history-length controls, and a nonrecurrent policy would help separate information effects from input-scale and architecture effects.

Signal saturation. The latent signal saturates near the Nash reference. Alternative nonsaturating maps and smaller, adaptive deviations are needed before interpreting small impulse responses as insensitivity to economically relevant departures.

Probe coverage. Testing upward price changes, best-response searches, multistep deviations, and other histories could reveal vulnerabilities missed by the current finite probe family.

Missing positive control. A functioning collusive positive control is missing. A known trigger policy under the analytical monitoring model should generate the predicted current-gain/continuation-loss tradeoff before the audit is described as a validated detector with known sensitivity. Proposition 1 verifies a negative-control implication, not the power of the entire procedure.

Model scope. All inference is conditional on one retained posterior and one stationary demand specification. Ten seeds characterize some training variability, not robustness to model misspecification. A confidence interval containing zero is not proof of no effect.

8.1 Reproducibility and Computational Accounting

All 20 PPO jobs are complete. The archived Python source hashes and final checkpoint hashes match their recorded identities. The manifest records maximum rank-normalized split $\hat{R}$ of 1.008945 and minimum bulk/tail ESS of 606.86, satisfying the software gates of 1.01 and 400. These diagnostics concern sampling quality (Vehtari et al., 2021). They do not validate Gaussianity or establish external market calibration. Independent reconstruction also requires the original calibration observations and the hash-matched posterior draws, not merely these diagnostic summaries.

The completed runs account for 20 million training interactions and $58{,}174{,}720$ total recorded interactions after diagnostics and audits. Recorded completed-job time totals about 44.06 hours. Three archived prior-attempt records remain; their costs are not covered by this completed-run total. Hardware-specific runtime comparisons require more telemetry. The broader plan contains SAC conditions, but no incomplete SAC comparison enters this manuscript. The PPO-only scope and its interpretation were chosen after inspecting results, so the paper is not a preregistered confirmatory test of a null effect.

8.2 Conclusion and Broader Impact

The theory identifies the continuation incentive needed to deter a profitable deviation. The experiment does not find strong evidence of that incentive in these PPO policies. Early high prices decline in both treatments. Observed deviation losses are mostly immediate. Representation differences do not establish a pricing mechanism. These are substantive boundaries on the proposed explanation. Information restrictions can also harm useful market learning or consumer comparison. This stationary experiment cannot determine their real-world welfare consequences or establish unlawful conduct. Prices are observable. Enforcement remains a hypothesis to test.

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AI Use Disclosure

Generative AI was used to assist with manuscript drafting and revision, mathematical derivation checks, implementation development and review, postprocessing of archived experimental outputs, and figure preparation. The numerical results reported here were computed from the archived simulator runs, not generated as hypothetical outcomes. AI-assisted exposition and proof checking do not replace author responsibility for the assumptions, derivations, code, references, and final claims.